

[CH 9] New Modes of Computing

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- In the mid 60s, a commercial computing installation was pretty standardized: a central IBM computer (or one from the other mainframe manufacturers) running Batch or Real-Time computing applications
- The role of the user was to feed data in and to interact with the machine when the application required it
- Computer manufacturing costs had fallen, performance per dollar had increased but the nature of data processing had barely changed
 - Applications were more sophisticated
 - Some batch operations became real-time
 - BUT computers were still delivered to naïve users by elite groups of software developers and system analysts
- The personal computer grew out of a totally different culture of computing

Time-Sharing

- Initially, computers in universities would be booked by someone for a given amount of time (half an hour, an hour, etc), during which they had exclusive access
 - This was wasteful because, of that time, only a few minutes of actual computing would occur
- Alternatively, there was batch processing. Multiple users' programs, which had been previously submitted to the computer center, would be run in a batch, and the results would be reported to the owners of the software
 - This made good use of the computer's time but terrible use of the programmers' / researchers' time
- 1959 **Christopher Strachey** proposed the idea of attaching several operating consoles (+ card reader & printer) to a single mainframe computer, so that each user would have the illusion that they were the only user
- MIT's John McCarthy proposed that the users would communicate through typewriter-like terminals
- MIT's time-sharing project, lead by **Robert Fano** and **Fernando Corbato** was the first of its kind.
- Named it CTSS (Compatible Time-Sharing System)
- First demonstrated in 1961
- Allowed 3 users to share the computer, letting them edit and correct programs and do other information-processing tasks
- This set the agenda for MIT and other universities for the next decade
- Within a year, universities, research organizations and manufacturers had begun to develop their own time-sharing systems

Dartmouth Basic

- The best known timesharing system
- Named the DTSS (Dartmouth Time Sharing System)
- CTSS was designed for computer scientists, but DTSS was intended for a much broader spectrum
- Design began 1962, when funding was secured for a time-sharing system by John Kemeny and Thomas E. Kurtz
- They ordered General Electric computers because the company had an interest in developing time sharing
- GE had made some hardware modifications in that sense but there was no software provided
- The computers were delivered in 1964 and during the spring both the operating system and the language called BASIC were developed
- The goal was to keep it so simple that it could be developed entirely by a group of students, which did happen
- BASIC = Beginners All-purpose Symbolic Instruction Code
- BASIC was designed so that undergraduates (liberal arts as well as science) could develop their own programs
- Previously, undergraduates generally had to use FORTRAN, which was not designed for education:
 - It was meant for scientists and engineers writing technical software
 - It was ugly and difficult

- It was slow
- The goals of BASIC were:
 - Be simple enough that one could program after only a few hours of lessons
 - Respond as expected to common formulas and constructions
 - Simple yet complete error messages for quick debugging without reading a manual
 - Permit extensions to carry the most sophisticated tasks required by experts
- Spring 1964 BASIC became available and in the next year it already began being taught to freshmen at Dartmouth
- It took just two 1-hour lectures to start programming
- Eventually most students learned to use the system
- 1968 it was already used by 23 schools and 10 New England colleges
- BASIC became the introductory language for all non-specialist students learning computing
- Every manufacturer had to provide a BASIC translator if it wanted to compete in the education market
- BASIC was riding the wave of time-sharing systems emerging in the late 1960s and early 1970s
- 1975 it was the first widely available language for the newly emerging personal computer
- It laid the foundations for Microsoft
- It may have been a technical step backwards, but it was a huge step forward in user-friendliness
- BASIC added a third group of computer user: the ones who developed their own programs and for whom computers were a personal information tool

J.C.R. Licklider and the Advanced Research Projects Agency (ARPA)

- ARPA began providing lavish funding in 1962 and it propelled time-sharing into the mainstream
- ARPA was established as a response to the Sputnik scare of October 1957 by President Dwight D. Eisenhower.
- ARPA was meant to sponsor and coordinate defense related research
- The requirement was not to deliver immediate results, but to pursue long-term payoff
- Among the projects was a \$7 million one for promoting the use of computers in defense, directed by MIT psychologist and computer scientist J.C.R. Licklider, who had been involved in human-factors design of radar display consoles at SAGE
- 1960 he published "Man-Computer Symbiosis", formulating the manifesto for human-computer interaction and reshaping computing for the next 20 years
- He advocated for the use of computers to augment human intellect, at odds with AI, where the dream was a computer that would rival the human intellect
- He believed computer scientists should develop systems that enabled people to use computers to enhance everyday work
- In ARPA he found the resources to pursue his vision
- He established the IPTO (Information Processing Techniques Office) within ARPA, which would exceed the funding of all other sources of US public computer research funding
- He encouraged his chosen university centers (MIT, Stanford, Carnegie-Mellon, University of Utah) to pursue time-sharing systems that would facilitate HCI studies
- 3 million USD went to MIT for the development of a leading-edge system named MAC
 - Based on IBM mainframes
 - Became operational in 1963
 - Built up to 160 consoles on campus and in the homes of faculty members
 - Up to 30 concurrent users
 - It could also be used to prepare and edit documents => paved the way for word processing systems
 - IBM and the major computer manufacturers remained unmoved by the project
- By 1965 MAC was fully operational and highly overloaded
- A new project, Multics, was launched (Multiplexed Information and Computing Service)
 - It was meant to be modular and support hundreds of users
 - Because IBM was technologically unsuited to time-sharing, MIT chose General Electric
 - GE was happy to include MIT in the specification process for its new Model 645 computer, dedicated for time-sharing
 - The software was developed by Bell Laboratories

- Pushed out of Multics, IBM realised that it would need to take time-sharing into consideration and announced the model 67 in 1966

The computer utility

- Time sharing was broadened into the concept of computer utility
- Originated in 1964 at MIT and was much alike a power utility
- An electricity user does not have a power generating plant, but draws electricity from a power utility
- Computer users could draw computing power from centralized mainframe utilities, instead of owning individual computers
- There were 2 main driving forces:
 - **Economic.** More interesting to computer manufacturers and buyers. Grosch's law states that the power of a computer varied as the square of its price. It would be much cheaper to have 25 users using a 200,000 USD computer than giving each of them their own 100,000 USD computer
 - **Visionary.** The aim was a democratization of computing, making it widely accessible
- The skeptics ended up being proven right, but did not stop the bandwagon
- By 1967 the time-sharing market was disputed by 20 firms
 - IBM offered the QUICKTRAN service for simple calculations in 5 cities
 - GE had time sharing services in 20 cities in the US
 - Licklider's old employer BBN had established a nationwide service called Telcomp
 - Tymshare (San Francisco)
 - Keydata (Boston)
 - Comshare (Ann Arbor, Michigan)
 - University Computing Company (UCC) was the biggest. It had major utilities in New York and Washington
 - By 1968 it had centers with terminals in 30 states and 12 countries
- By the end of the 60s, they had expanded across North America and into Europe
- Almost overnight, the bottom dropped out
- 1970 criticism started
- 1971 several firms were in trouble
 - UCC share price dove from 186 to 17 USD in a few months
- The causes:
 - The software crisis
 - Big computer utilities were reliant on software
 - Falling hardware prices
- The Multics project did not go anywhere near as smoothly as the Dartmouth Basic project
 - After years of struggles, Bell Labs pulled out in 1969
 - 1970 GE withdrew the model 645
 - A few months later it pulled out of the mainframe business altogether
- Computer utility was a failure

UNIX

From the Multics disaster came UNIX

- Developed at bell labs during 1969-1974
- Avoided complexity through clear minimalist design
- Developers Ken Thompson and Dennis M. Ritchie managed to convince the Bell Labs patent department to use a UNIX based text processor to develop patent specifications => they got funding for a better computer and they chose the PDP-11/45
- The system was rewritten in the language "C"
 - C was designed by Ritchie for the purpose
 - It was a systems implementation language
 - Designed for writing programming systems
 - It made UNIX portable, it could be installed on any computer system (unique at the time for an OS)

- Many wanted smaller independent computers, but there was no suitable OS from manufacturers
- UNIX filled this void
- From 1974 more and more universities started using UNIX
- Starting 1977 UNIX started growing with more and more additions
 - The design permitted this organic growth
 - UNIX was on the path to becoming the standard OS for the 1980s
- 1983 Thompson and Ritchie received the A.M. Turing award
- In the late 1990s, UNIX took another mantle on in the shape of Linux
 - Started by Finnish undergraduate computer scientists Linus Torvalds in 1992, it became one of the centerpieces of "open-source" software

From MIT to Minicomputers

- The introduction of integrated circuit electronics reduced the cost of computer hardware dramatically (factor of 100) => no more justification for time-sharing
- The minicomputer did not emerge from the established computer industry
- It was a computer born of MIT and the East Coast microelectronics industry
- One of the most important companies is DEC (Digital Equipment Corporation), founded by Kenneth Olsen and Harlan Anderson 1957
- For the first few years it produced digital circuit boards to fund its first computer development
- The first computer was the PDP-1 in 1960
 - Cost an unbelievable 125,000 USD for a basic model
 - The price was because the PDP was aimed at the science and engineering market, not the data processing one
 - Scientists and engineers did not need advanced peripherals, wrote their own software and did not need sales people to analyze their applications. This meant DEC could reduce costs drastically
- PDP-1 and the next 6 models were a modest success
- PDP-8 came out in 1965 and exploited the new integrated circuits => far smaller and cheaper
 - It fit in a packaging case and cost 18,000 USD
 - It was an instant success and sold hundreds of models in one year
 - The cheap price made them more affordable.
 - Many people got their first taste of hands-on computing
 - Some got attached to their PDP-8's, calling them their personal computers
- This inspired students, young technicians and even seasoned engineers and a hobbyist culture appeared
- By 1969 several companies were competing with DEC in the Minicomputer industry (Data General, Prime Computer, Hewlett-Packard, Harris, Honeywell)
- DEC kept the lead and by 1970 it was the 3rd biggest computer manufacturer worldwide after IBM and Sperry Rand's UNIVAC

Semiconductors and Silicon Valley

- By the 1960s the place for innovation was Silicon Valley in California
- At first it was semiconductor manufacturers like Shockley Semiconductor (Shockley had invented the transistor) and Fairchild Semiconductor (founded by Robert Noyce)
- The area was close to Stanford
- Fairchild Semiconductor co-founder Jean Hoerni developed the planar process, allowing multiple transistors to be deposited on a single silicon wafer
- The silicon wafers gave the area its name: Silicon Valley
- Just like Fairchild was a spin-off of Shockley Semiconductor, so it too spun-off startups who spun-off startups of their own
- Half of the 85 major semiconductor manufacturers in the US originated from Fairchild
 - Intel
 - AMD
 - National Semiconductor

- Intel was the second startup founded by Robert Noyce, along with Gordon Moore
- Intel incorporated in 1968 and within 2 years it was profitably producing integrated-circuit memory chips
- Within 5 years it was worth \$66 million and employed 2,500 people
- Silicon Valley companies prided themselves on being fast, flexible and meritocratic
- The number of components per chip was doubling each year. By 1970 there were chips with 1000 components
- Moore's Law stated that the number of active components per integrated circuit would double about every 18 months

Chips in Consumer Products

- As performance went up, prices went down simultaneously
- This drove firms to innovate by improving existing product lines and creating new markets
 - Ex. Calculators. Integrated circuits reduced the cost dramatically, to below \$5.
- The most significant user of high-density integrated circuits was the video game industry
- 1971 Nolan Bushnell founded Atari, with the first product being Pong
 - When prices dropped, Atari started making high-volume home versions of Pong that sold for \$350
 - This was followed by a string of games, with prices dropping to \$50 and sales climbing to \$40 million by 1976
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